Input: syndrome $\vec{s}$
$\vec{s} = [\ 0,\ \ 1, \ldots, 1,\ \ 0\ ]^T$
matrix product state
parallel contraction
$O(M\chi^2)$
MPS with logical indices
trace other indices to get marginal probability
$P(l_1 = 1|\vec{s})$
$P(l_2 = 1|\vec{s})$
$P(l_K = 1|\vec{s})$
$O(K\chi^4)$
0.78
0.52
0.12
0
0.4
0.6
1
0
1
$l_1 = 1$
l_2 unknown
$l_K = 1$
get conditional probability for unknown indices
$P(l_2 = 1|l_1 = 1, l_K = 0, \vec{s})$
$P(l_3 = 1|l_1 = 1, l_K = 0, \vec{s})$
1
1
0
repeat utill all indices are known
Output: most likely logical operator $\vec{l}$